

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version D — SOLUTIONS

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About Version D. Generated with GPT-6 Astra in Codex. This set targets about thirty minutes of work in a fifty-minute period; it has not been timed with students. Coverage: §§0.1–0.3, §§1.1–1.6, §2.1, and §§2.3–2.5. Please report errors, ambiguity, out-of-coverage questions, or timing problems to Professor Dupuy.

Problem 1. (30 points) **True or false; prove or disprove.**

- (a) **TRUE.** If $f(a) = f(b)$, then $g(f(a)) = g(f(b))$; injectivity of $g \circ f$ implies $a = b$.
- (b) **TRUE.** Reflexivity follows from $2a$ being even; symmetry from $a + b = b + a$; transitivity from $a + c = (a + b) + (b + c) - 2b$ being even when $a + b$ and $b + c$ are even.
- (c) **FALSE.** The symmetric, invertible matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{have} \quad AB = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix},$$

which is not symmetric, so the set is not closed under multiplication.

- (d) **FALSE.** Since $j^{-1} = -j$, one has $ij^{-1} = -ij = -k \neq k$.
- (e) **TRUE.** If $q \neq 0$ is rational and $n > 0$ is an integer, then $nq \neq 0$, so no positive multiple of q is the additive identity.
- (f) **FALSE.** In the multiplicative group $\{1, -1\}$, $a = -1$ satisfies $a^6 = 1$ but $a^3 = -1 \neq 1$.

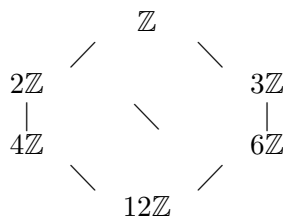
Grading note. Each item is worth 2 points for the correct T/F choice and 3 for a valid reason. In (b), the justification should cover all three equivalence-relation properties, or identify this as equality of parity. In (c), the proposed factors must both be symmetric and invertible, and their product must fail symmetry. Any valid counterexample earns full credit in (c) or (f). In (e), the group operation is addition.

Problem 2. (25 points) **Subgroups of the integers.**

- (a) (8 points) Both 28 and 42 belong to the subgroup $14\mathbb{Z}$, so $\langle 28, 42 \rangle \subseteq 14\mathbb{Z}$. Conversely, $14 = 42 - 28 \in \langle 28, 42 \rangle$. A subgroup containing 14 contains every integer multiple of 14, hence $14\mathbb{Z} \subseteq \langle 28, 42 \rangle$.
- (b) (6 points) An integer belongs to $4\mathbb{Z} \cap 6\mathbb{Z}$ precisely when both 4 and 6 divide it. This is equivalent to divisibility by $\text{lcm}(4, 6) = 12$, so $4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}$.
- (c) (11 points) Every subgroup of $\mathbb{Z}$ is $d\mathbb{Z}$ for a unique nonnegative integer d . Here it contains $12 \neq 0$, so $d > 0$, and $12\mathbb{Z} \subseteq d\mathbb{Z}$ holds exactly when $d \mid 12$. Thus the complete list is

$$\mathbb{Z}, \quad 2\mathbb{Z}, \quad 3\mathbb{Z}, \quad 4\mathbb{Z}, \quad 6\mathbb{Z}, \quad 12\mathbb{Z}.$$

The lattice is



For precision, its seven edges (larger subgroup first) are

$$(\mathbb{Z}, 2\mathbb{Z}), (\mathbb{Z}, 3\mathbb{Z}), (2\mathbb{Z}, 4\mathbb{Z}), (2\mathbb{Z}, 6\mathbb{Z}), \\ (3\mathbb{Z}, 6\mathbb{Z}), (4\mathbb{Z}, 12\mathbb{Z}), (6\mathbb{Z}, 12\mathbb{Z}).$$

Recall that $a\mathbb{Z} \subseteq b\mathbb{Z}$ exactly when $b \mid a$; containment reverses divisibility of the positive generators.

Grading note. In (a), award 4 points per inclusion; finding $42 - 28 = 14$ alone establishes only one direction. In (b), award 3 for $12\mathbb{Z}$ and 3 for the common-multiple justification. In (c), award 3 for the six subgroups, 4 for the seven correct lattice edges, and 4 for completeness using the classification of subgroups of $\mathbb{Z}$ and $d \mid 12$. A reversed diagram loses the placement credit but not credit for a correct list or proof.

Problem 3. (25 points) **A rational matrix group.**

(a) (8 points) Direct multiplication gives

$$U(s)U(t) = \begin{pmatrix} 1 & s+t \\ 0 & 1 \end{pmatrix} = U(s+t), \quad U(t)^{-1} = U(-t) = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}.$$

The identity is $U(0)$, and rational numbers are closed under addition and negation. Thus H contains the identity and is closed under products and inverses, so it is a subgroup of $\mathrm{GL}_2(\mathbb{Q})$.

(b) (9 points) For $s, t \in \mathbb{Q}$,

$$\varphi(s+t) = U(s+t) = U(s)U(t) = \varphi(s)\varphi(t),$$

so φ is a homomorphism. If $U(s) = U(t)$, comparison of the upper-right entries gives $s = t$, so it is injective. Every member of H is $U(t)$ for some $t \in \mathbb{Q}$ by definition, so it is surjective. Therefore it is an isomorphism.

(c) (8 points) The formulas in (a) imply $U(q)^n = U(nq)$ for every integer n . If $q = 0$, then $\langle U(q) \rangle = \{U(0)\}$, which misses $U(1)$. If $q \neq 0$, then $U(q/2) \in H$, but

$$U(q/2) = U(q)^n = U(nq) \implies q/2 = nq \implies n = 1/2,$$

which is impossible for an integer n . Thus no element of H generates H .

Grading note. In (a), award 3 points for the product, 2 for the inverse, and 3 for the subgroup argument including the identity. In (b), award 3 each for the homomorphism property, injectivity, and surjectivity. In (c), award 2 for describing all integer powers, 1 for excluding $q = 0$, and 5 for the contradiction using $q/2$. Infinitude alone does not prove noncyclicity: $\mathbb{Z}$ is an infinite cyclic group.

Problem 4. (20 points) **Homomorphisms into S_3 .**

- (a) (8 points) If f is a homomorphism and $y = f(x)$, then $y^6 = f(x^6) = f(1) = 1$, and necessarily $f(x^m) = y^m$ for all integers m . Conversely, suppose $y^6 = 1$. If $x^m = x^n$, then $6 \mid (m - n)$ because x has order 6. Write $m - n = 6k$; then $y^m = y^n(y^6)^k = y^n$. Thus the rule $f(x^m) = y^m$ is well defined. It is a homomorphism because

$$f(x^m x^n) = y^{m+n} = y^m y^n = f(x^m) f(x^n).$$

This also proves uniqueness for each chosen y .

- (b) (7 points) The six elements of S_3 are

$$1, \quad (12), \quad (13), \quad (23), \quad (123), \quad (132).$$

Their orders are respectively 1, 2, 2, 2, 3, 3, all divisors of 6, so each satisfies $y^6 = 1$. By (a), there are exactly **six** homomorphisms; different choices of $f(x)$ give different maps.

- (c) (5 points) None is injective. For any choice $y = f(x)$ above, let $d = |y| \in \{1, 2, 3\}$. Then $f(x^d) = y^d = 1 = f(1)$, but $x^d \neq 1$ because x has order 6 and $0 < d < 6$. Thus distinct elements of C have the same image.

Grading note. In (a), award 2 points for necessity, 3 for checking well-definedness under changes of exponent, and 3 for the homomorphism property and uniqueness. In (b), award 3 for the complete list, 2 for checking $y^6 = 1$, and 2 for the count via (a). In (c), award 2 for the answer and 3 for a valid reason, such as the displayed collision or the fact that an injective homomorphism preserves element orders and S_3 has no element of order 6. Equal group sizes alone do not imply that a particular homomorphism is injective.